

problemsetup

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Contents

1	Set up file containing data specific to the current problem	1
2	References	38
3	INDEX	39

1 Set up file containing data specific to the current problem

\$Id: problemsetup.web,v 1.37 2009/11/25 20:01:36 dstotler Exp \$

There are two levels of physics input to DEGAS 2. The input files set up by *datasetup* comprise the reference level: in practice, the sum total of the data available to the code (although in principle it could be smaller). The second level is the problem level of data. This prescribes the species, reactions, materials, and PMI which will serve as the physical model to be used in carrying out the simulation at hand. In some parts of the (internal) code, these are also referred to as the subset data since they represent a subset of the reference data.

At the problem level the concepts of test species and background species are introduced. Most simply, test species are the ones DEGAS 2 will track as they collide off of background species. The use of the word species here is important: both of the test and background lists are subsets of the “species” list. One more precise distinction between the two species types is that we assume that we know the distribution function (in space and velocity) of the background species; in fact, such information is required input to DEGAS 2. On the other hand, we are attempting to *compute* the test species distribution function, moments of which serve as the primary output of DEGAS 2.

The problem input file associated with the symbolic name in **problem_infile** in **degas2.in** consists of five sections, one for each of the problem-level physics subsets, separated by the corresponding heading (in capital letters). It is the object symbols (as opposed to the longer “names”) which are used in these lists. For example,

TEST

0 D D2 D2+

BACKGROUND

e D+

REACTION

hchex
hionize
h2dis
h2ion
h2dision
h2pdision
h2pdis
h2pdisrec

MATERIALS

mo
mirror

PMI

dreflmo

```
hdesorbmo
h2desorbmo
hmirror
h2mirror
```

This file is read by *problemsetup*. The reference lists are searched for these symbols. Associations between the subset lists which are needed by the code for tracking the test species are built up. The required atomic and surface data are read in and repackaged for efficient run-time use. All of the resulting information is written to the netCDF problem file, which corresponds to the symbolic name *problemfile* in *degas2.in*.

```
"problemsetup.f" 1 ≡
  @m FILE 'problemsetup.web'
```

The unnamed module.

```
"problemsetup.f" 1.1 ≡
  program problemsetup
    implicit none_f77
    implicit none_f90

    @#if SUN ^ FORTRAN77
      call float_abort
    @#endif

    call readfilenames
    call read_problem

    call init_var0_list

    call init_reaction
    call init_pmi

    call finish_var0_list

    call nc_write_problem

    stop
  end
```

⟨ Functions and Subroutines 1.2 ⟩

Read in data from **problem_infile**.

⟨Functions and Subroutines 1.2⟩ ≡

```

subroutine read_problem
  implicit none
  pr_common
  rf_common
  implicit none

  integer length // Local
  character*LINELEN line
  character*FILELEN tempfile

  ⟨Memory allocation interface 0⟩ // Common
  st_decls

  problem_version = 'unknown'
  tempfile = filenames_array_problem_infile
  assert(tempfile ≠ char_undef)
  open(unit = disk_in, file = tempfile, form = 'formatted', status = 'old')
  assert(read_string(disk_in, line, length))
  assert(length ≤ len(line))
  problem_version = line(:length)

  call init_problem
  if (¬read_string(disk_in, line, length)) then
    line = 'END'
  end if

loop1: continue
  if (line ≡ 'END')
    goto eof
  if (line ≡ 'TEST') then
    call read_test(disk_in, line)
  else if (line ≡ 'BACKGROUND') then
    call read_background(disk_in, line)
  else if (line ≡ 'REACTION') then
    call read_reaction(disk_in, line)
  else if (line ≡ 'MATERIALS') then
    call read_materials(disk_in, line)
  else if (line ≡ 'PMI') then
    call read_pmi(disk_in, line)
  end if
  goto loop1
eof: continue
  call check_problem

  return
end

```

See also sections 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 1.10, 1.11, 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.19, 1.20, 1.21, 1.22, 1.23, 1.24, 1.25, and 1.26.

This code is used in section 1.1.

Initialize problem.

⟨Functions and Subroutines 1.2⟩ +≡

```

subroutine init_problem
  implicit none f77
  pr_common      // Common
  sp_common
  ps_common
  implicit none f90
  integer i      // Local

  ⟨Memory allocation interface 0⟩

  call nc_read_elements
  call nc_read_species
  call nc_read_reactions
  call nc_read_materials
  call nc_read_pmi

  pr_test_num = 0
  pr_background_num = 0
  pr_reaction_num = 0
  pr_materials_num = 0
  pr_pmi_num = 0

  num_generics = 0
  var_alloc(generics)
  do i = 1, sp_num
    genericsi = 0
  end do

  return
end

```

Read in test particle labels; make required entries in generic species arrays.

⟨ Functions and Subroutines 1.2 ⟩ +=

```

subroutine read_test(unit, line)
  implicit none f77
  pr_common      // Common
  sp_common
  ps_common
  implicit none f90
  integer unit      // Input
  character*(*) line  // Output
  integer length, p, b, e, i  // Local

  ⟨ Memory allocation interface 0 ⟩
  st_decls

  var_alloc(problem_species_test)
  do i = 1, sp_num
    problem_species_testi = 0
  end do

loop1: continue
  if (¬read_string(unit, line, length))
    line = 'END'
  if (line ≡ 'END' ∨ line ≡ 'TEST' ∨ line ≡ 'BACKGROUND' ∨ line ≡ 'REACTION' ∨ line ≡
    'MATERIALS' ∨ line ≡ 'PMI' ∨ line ≡ 'SCORES')
    goto eof
  assert(length ≤ len(line))
  length = parse_string(line( : length))

  p = 0
loop2: continue
  if (¬next_token(line( : length), b, e, p))
    goto loop1
  pr_test_num++
  var_realloca(problem_test_sp)
  pr_test(pr_test_num) = sp_lookup(line(b : e))
  assert(pr_test(pr_test_num) > 0)
  problem_species_test(pr_test(pr_test_num)) = pr_test_num

  if (genericssp_generic(pr_test(pr_test_num)) ≡ 0) then    /* First appearance of this generic species;
    update num_generics and generics. In case this entry in the test list is not the archetype
    generic of its isotopic family (i.e., the value of sp_generic), need to also enter num_generics in
    its slot in the generics array. E.g., if D2 were in a problem, but the archetype generic H2 were
    not. */
    num_generics++
    genericssp_generic(pr_test(pr_test_num)) = num_generics

    if (pr_test(pr_test_num) ≠ sp_generic(pr_test(pr_test_num)))
      genericspr_test(pr_test_num) = num_generics

  else    /* This entry is equivalent to a generic species which has already been added to the generics
    array. */
    genericspr_test(pr_test_num) = genericssp_generic(pr_test(pr_test_num))
  end if

  goto loop2

```

§1.4 [#5] **problemsetup** Set up file containing data specific to the current problem 6

```
eof: return  
     end
```

Read in background particle labels; make required entries in generic species arrays.

⟨Functions and Subroutines 1.2⟩ +=

```

subroutine read_background(unit, line)
  implicit none
  pr_common      // Common
  sp_common
  ps_common
  implicit none
  integer unit      // Input
  character*(*) line  // Output
  integer length, p, b, e, i  // Local

  ⟨Memory allocation interface 0⟩
  st_decls

  var_alloc(problem_species_background)
  do i = 1, sp_num
    problem_species_backgroundi = 0
  end do

loop1: continue
  if (¬read_string(unit, line, length))
    line = 'END'
  if (line ≡ 'END' ∨ line ≡ 'TEST' ∨ line ≡ 'BACKGROUND' ∨ line ≡ 'REACTION' ∨ line ≡
    'MATERIALS' ∨ line ≡ 'PMI' ∨ line ≡ 'SCORES')
    goto eof
  assert(length ≤ len(line))
  length = parse_string(line)

  p = 0
loop2: continue
  if (¬next_token(line(: length), b, e, p))
    goto loop1
  pr_background_num++
  var_realloca(problem_background_sp)
  pr_background(pr_background_num) = sp_lookup(line(b : e))
  assert(pr_background(pr_background_num) > 0)
  problem_species_background(pr_background(pr_background_num)) = pr_background_num

  /* Handle generics just as we did for test species: */
  if (genericssp_generic(pr_background(pr_background_num)) ≡ 0) then
    num_generics++
    genericssp_generic(pr_background(pr_background_num)) = num_generics
    if (pr_background(pr_background_num) ≠ sp_generic(pr_background(pr_background_num)))
      genericspr_background(pr_background_num) = num_generics
    else
      genericspr_background(pr_background_num) = genericssp_generic(pr_background(pr_background_num))
    end if
  goto loop2

eof: return
end

```

Read in reaction particle labels.

⟨ Functions and Subroutines 1.2 ⟩ +=

```

subroutine read_reaction(unit, line)
  implicit none f77
  pr_common     // Common
  rc_common
  implicit none f90
  integer unit     // Input
  character*(*) line     // Output
  integer length, p, b, e     // Local

  ⟨ Memory allocation interface 0 ⟩
  st_decls

  pr_reaction_dim = 0
  assert(pr_reaction_num ≡ 0)
loop1: continue
  if (¬read_string(unit, line, length))
    line = 'END'
  if (line ≡ 'END' ∨ line ≡ 'TEST' ∨ line ≡ 'BACKGROUND' ∨ line ≡ 'REACTION' ∨ line ≡
    'MATERIALS' ∨ line ≡ 'PMI' ∨ line ≡ 'SCORES')
    goto eof
  assert(length ≤ len(line))
  length = parse_string(line)

  p = 0
loop2: continue
  if (¬next_token(line(: length), b, e, p))
    goto loop1
  pr_reaction_dim++
  var_realloca(problem_rc)
  pr_reaction(pr_reaction_dim) = rc_lookup(line(b : e))
  assert(pr_reaction(pr_reaction_dim) > 0)
  goto loop2
eof: continue

  pr_reaction_num = pr_reaction_dim
  if (pr_reaction_dim ≡ 0) then
    pr_reaction_dim = 1
    var_realloca(problem_rc)
  end if

  return
end

```

Read in materials labels.

⟨ Functions and Subroutines 1.2 ⟩ +=

```

subroutine read_materials(unit, line)
  implicit none f77
  pr_common      // Common
  ma_common
  implicit none f90
  integer unit      // Input
  character*(*) line  // Output
  integer length, p, b, e, i  // Local

  ⟨ Memory allocation interface 0 ⟩
  st_decls

  var_alloc(problem_materials_sub)
  do i = 1, ma_num
    problem_materials_subi = 0
  end do

loop1: continue
  if (¬read_string(unit, line, length))
    line = 'END'
  if (line ≡ 'END' ∨ line ≡ 'TEST' ∨ line ≡ 'BACKGROUND' ∨ line ≡ 'REACTION' ∨ line ≡
    'MATERIALS' ∨ line ≡ 'PMI' ∨ line ≡ 'SCORES')
    goto eof

  assert(length ≤ len(line))
  length = parse_string(line)

  p = 0
loop2: continue
  if (¬next_token(line(: length), b, e, p))
    goto loop1
  pr_materials_num++
  var_realloca(problem_materials_ref)
  pr_mat_ref(pr_materials_num) = ma_lookup(line(b : e))
  assert(pr_mat_ref(pr_materials_num) > 0)
  problem_materials_sub(pr_mat_ref(pr_materials_num)) = pr_materials_num

  goto loop2

eof: return
end

```

Read in plasma-materials interaction labels.

```

⟨ Functions and Subroutines 1.2 ⟩ +=
subroutine read_pmi(unit, line)
  implicit none f77
  pr_common      // Common
  pm_common
  implicit none f90
  integer unit      // Input
  character*(*) line // Output
  integer length, p, b, e, i // Local

  ⟨ Memory allocation interface 0 ⟩
  st_decls

  var_alloc(problem_pmi_sub)
  do i = 1, pm_num
    problem_pmi_subi = 0
  end do

loop1: continue
  if (¬read_string(unit, line, length))
    line = 'END'
  if (line ≡ 'END' ∨ line ≡ 'TEST' ∨ line ≡ 'BACKGROUND' ∨ line ≡ 'REACTION' ∨ line ≡
    'MATERIALS' ∨ line ≡ 'PMI' ∨ line ≡ 'SCORES')
    goto eof
  assert(length ≤ len(line))
  length = parse_string(line)

  p = 0
loop2: continue
  if (¬next_token(line(: length), b, e, p))
    goto loop1
  pr_pmi_num++
  var_realloca(problem_pmi_ref)
  pr_pm_ref(pr_pmi_num) = pm_lookup(line(b : e))
  assert(pr_pm_ref(pr_pmi_num) > 0)
  problem_pmi_sub(pr_pm_ref(pr_pmi_num)) = pr_pmi_num

  goto loop2

eof: return
end

```

Setup initial entries in list of dependent variables. This routine does two of the three “sections” of this array: hardwired (entries corresponding to macros) and background terms (repeated for each background species). The reaction-specific entries will be set in *xs_copy* as needed.

⟨ Functions and Subroutines 1.2 ⟩ +=

subroutine *init_var0_list*

implicit none f77

pr_common // Common

sp_common

implicit none f90

integer *i* // Local

⟨ Memory allocation interface 0 ⟩

pr_var0_num = 1 // Will force *mem_alloc* to be called.

var_alloc(*pr_var0_list*)

pr_var0_num = *pr_var_mass_change* - 1 + *pr_num_change_vars* * (*pr_background_num* + *pr_test_num* + 1)

 // The last “1” here accounts for the generic entries.

@#if 0

var_realloc(*pr_var0_list*) // Need here since will be using later also

@#endif

var_realloc(*pr_var0_list*, 1, *mem_size*(*pr_var0_num*))

do *i* = 1, *pr_var0_num*

*pr_var0_list*_{*i*} = *char_uninit*

end do

*pr_var0_list*_{*pr_var_unknown*} = 'unknown'

*pr_var0_list*_{*pr_var_mass*} = 'mass'

*pr_var0_list*_{*pr_var_momentum_vector*} = 'momentum_vector'

*pr_var0_list*_{*pr_var_momentum_2*} = 'momentum_2'

*pr_var0_list*_{*pr_var_momentum_3*} = 'momentum_3'

*pr_var0_list*_{*pr_var_energy*} = 'energy'

*pr_var0_list*_{*pr_var_angle*} = 'angle'

*pr_var0_list*_{*pr_var_emitter_v_vector*} = 'emitter_v_vector'

*pr_var0_list*_{*pr_var_emitter_v_2*} = 'emitter_v_2'

*pr_var0_list*_{*pr_var_emitter_v_3*} = 'emitter_v_3'

*pr_var0_list*_{*pr_var_emitter_vf_Maxwell_vector*} = 'emitter_vf_Maxwell_vector'

*pr_var0_list*_{*pr_var_emitter_vf_Maxwell_2*} = 'emitter_vf_Maxwell_2'

*pr_var0_list*_{*pr_var_emitter_vf_Maxwell_3*} = 'emitter_vf_Maxwell_3'

*pr_var0_list*_{*pr_var_emitter_vth_Maxwell*} = 'emitter_vth_Maxwell'

*pr_var0_list*_{*pr_var_xz_stress*} = 'xz_stress' // For Couette example

*pr_var0_list*_{*pr_var_mass_change*} = 'mass_change' // Generic entries

*pr_var0_list*_{*pr_var_momentum_change_vector*} = 'momentum_change_vector'

*pr_var0_list*_{*pr_var_momentum_change_2*} = 'momentum_change_2'

*pr_var0_list*_{*pr_var_momentum_change_3*} = 'momentum_change_3'

*pr_var0_list*_{*pr_var_energy_change*} = 'energy_change'

*pr_var0_list*_{*pr_var_mass_in*} = 'mass_in'

*pr_var0_list*_{*pr_var_momentum_in_vector*} = 'momentum_in_vector'

*pr_var0_list*_{*pr_var_momentum_in_2*} = 'momentum_in_2'

*pr_var0_list*_{*pr_var_momentum_in_3*} = 'momentum_in_3'

*pr_var0_list*_{*pr_var_energy_in*} = 'energy_in'

*pr_var0_list*_{*pr_var_mass_out*} = 'mass_out'

*pr_var0_list*_{*pr_var_momentum_out_vector*} = 'momentum_out_vector'

*pr_var0_list*_{*pr_var_momentum_out_2*} = 'momentum_out_2'

```

pr_var0_list pr_var_momentum_out_3 = 'momentum_out_3'
pr_var0_list pr_var_energy_out = 'energy_out'

assert(pr_background_num ≥ 1)
do i = 1, pr_background_num
  pr_var0_list pr_var_problem_sp_index(pr_var_mass_change, pr_problem_sp_back(i)) = 'back_mass_change_' ||
    sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_change_vector, pr_problem_sp_back(i)) =
    'back_momentum_change_vector_' || sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_change_2, pr_problem_sp_back(i)) =
    'back_momentum_change_2_' || sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_change_3, pr_problem_sp_back(i)) =
    'back_momentum_change_3_' || sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_energy_change, pr_problem_sp_back(i)) =
    'back_energy_change_' || sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_mass_in, pr_problem_sp_back(i)) = 'back_mass_in_' ||
    sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_in_vector, pr_problem_sp_back(i)) =
    'back_momentum_in_vector_' || sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_in_2, pr_problem_sp_back(i)) =
    'back_momentum_in_2_' || sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_in_3, pr_problem_sp_back(i)) =
    'back_momentum_in_3_' || sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_energy_in, pr_problem_sp_back(i)) = 'back_energy_in_' ||
    sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_mass_out, pr_problem_sp_back(i)) = 'back_mass_out_' ||
    sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_out_vector, pr_problem_sp_back(i)) =
    'back_momentum_out_vector_' || sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_out_2, pr_problem_sp_back(i)) =
    'back_momentum_out_2_' || sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_out_3, pr_problem_sp_back(i)) =
    'back_momentum_out_3_' || sp_sy(pr_background(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_energy_out, pr_problem_sp_back(i)) = 'back_energy_out_' ||
    sp_sy(pr_background(i))
end do
assert(pr_test_num ≥ 1)
do i = 1, pr_test_num
  pr_var0_list pr_var_problem_sp_index(pr_var_mass_change, pr_problem_sp_test(i)) = 'test_mass_change_' ||
    sp_sy(pr_test(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_change_vector, pr_problem_sp_test(i)) =
    'test_momentum_change_vector_' || sp_sy(pr_test(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_change_2, pr_problem_sp_test(i)) =
    'test_momentum_change_2_' || sp_sy(pr_test(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_change_3, pr_problem_sp_test(i)) =
    'test_momentum_change_3_' || sp_sy(pr_test(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_energy_change, pr_problem_sp_test(i)) =
    'test_energy_change_' || sp_sy(pr_test(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_mass_in, pr_problem_sp_test(i)) = 'test_mass_in_' ||
    sp_sy(pr_test(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_in_vector, pr_problem_sp_test(i)) =
    'test_momentum_in_vector_' || sp_sy(pr_test(i))
  pr_var0_list pr_var_problem_sp_index(pr_var_momentum_in_2, pr_problem_sp_test(i)) =

```

```

    'test_momentum_in_2_' || sp_sy(pr_test(i))
pr_var0_list pr_var_problem_sp_index(pr_var_momentum_in_3, pr_problem_sp_test(i)) =
    'test_momentum_in_3_' || sp_sy(pr_test(i))
pr_var0_list pr_var_problem_sp_index(pr_var_energy_in, pr_problem_sp_test(i)) = 'test_energy_in_' ||
    sp_sy(pr_test(i))
pr_var0_list pr_var_problem_sp_index(pr_var_mass_out, pr_problem_sp_test(i)) = 'test_mass_out_' ||
    sp_sy(pr_test(i))
pr_var0_list pr_var_problem_sp_index(pr_var_momentum_out_vector, pr_problem_sp_test(i)) =
    'test_momentum_out_vector_' || sp_sy(pr_test(i))
pr_var0_list pr_var_problem_sp_index(pr_var_momentum_out_2, pr_problem_sp_test(i)) =
    'test_momentum_out_2_' || sp_sy(pr_test(i))
pr_var0_list pr_var_problem_sp_index(pr_var_momentum_out_3, pr_problem_sp_test(i)) =
    'test_momentum_out_3_' || sp_sy(pr_test(i))
pr_var0_list pr_var_problem_sp_index(pr_var_energy_out, pr_problem_sp_test(i)) = 'test_energy_out_' ||
    sp_sy(pr_test(i))
end do

return
end

```

Finish assignments to the list of dependent variable labels. This routine will serve to make an additional assignments to *pr_var0_list* beyond those made by *xs_copy* and to clean up its memory allocation.

The only task carried out here at present is to set up additional emission rate variables corresponding to each wavelength. Here's the long-winded explanation of why this is needed: "In general" the emission rate and wavelength variables should both have the name of the line appended to the variable names. For the case of greatest interest, this would entail carrying around 3 sets of (usually) identical data - one for each hydrogen isotope. Alternatively, the problem is that we're saying "all of the atomic physics for the hydrogen isotopes is the same except the wavelength"; the fact that there is an exception strictly speaking invalidates the notion of generic species. However, the other payoffs of generic species heavily outweigh this one exception. Here, the data files carry a generic name and multiple wavelengths. Each of these should now be in the *pr_var0_list*. However, the scoring dependent variables will refer to emission rates for explicit lines. We define here those additional emission rates, one for each wavelength. The individual reaction-processing subroutines will need to make the connection between the generic emission rate variable and the specific one based on the computed value of the wavelength. If needed, this procedure could be easily extended to handle other wavelength-dependent data.

```
"problemsetup.f" 1.10 ≡
  @m max_lines 3

⟨ Functions and Subroutines 1.2 ⟩ +=
  subroutine finish_var0_list

    implicit_none_f77
    pr_common
    implicit_none_f90

    integer i, i_e_rate, num_lines, old_num    // Local
    character*pr_tag_string_length line, lines_max_lines
    st_decls

    ⟨ Memory allocation interface 0 ⟩

    num_lines = 0
    old_num = pr_var0_num
    do i = 1, pr_var0_num
      if (pr_var0_list_iSP(1:10) ≡ 'wavelength') then
        line = pr_var0_list_iSP(11:)
        i_e_rate = string_lookup('emission_rate' || line, pr_var0_list, pr_var0_num)
        if (i_e_rate ≡ 0) then    // Add to list if not there already
          num_lines++
          pr_var0_num++
          var_realloca(pr_var0_list)
          lines_num_lines = line
        end if
      end if
    end do

    if (num_lines > 0) then
      @#if 0
        var_realloc(pr_var0_list, pr_var0_num, pr_var0_num + num_lines)
      @#endif
      @#if 0
        pr_var0_num += num_lines
        var_realloca(pr_var0_list)
```

```
@#endif
  do  $i = 1, num\_lines$ 
     $pr\_var0\_list_{old\_num+i} = 'emission\_rate' \parallel lines_i$ 
  end do
end if

 $var\_reallocb(pr\_var0\_list)$ 

return
end
```

Check problem specification. This routine sets up the arrays which will actually be used to implement reactions and PMI in the run. This involves not only matching up reactions and PMI with the test species, but also allowing for the isotopically equivalent combinations.

⟨ Functions and Subroutines 1.2 ⟩ +≡

```

subroutine check_problem
  implicit none f77
  pr_common     // Common
  sp_common
  rc_common
  ma_common
  pm_common
  ps_common
  implicit none f90
  integer i, j, k, l, m, ts_reagent, bk_reagent     // Local

  pr_test_decl(test)
  pr_background_decl(background)
  pr_generic_decl(generic)
  pr_generic_decl(generic_reagentspr_bkrc_reagent_max)
  pr_background_decl(backpr_bkrc_reagent_max)

  sp_decl(reagentsrc_reagent_max)

  ⟨ Memory allocation interface 0 ⟩
  st_decls

  var_reallocb(problem_test_sp)
  var_reallocb(problem_background_sp)
  var_reallocb(problem_rc)
  var_reallocb(problem_materials_ref)
  var_reallocb(problem_pmi_ref)

  write(stdout, *) 'Number_of:'
  write(stdout, *) '  test_species= ', pr_test_num
  write(stdout, *) '  background_species= ', pr_background_num
  write(stdout, *) '  materials= ', pr_materials_num
  write(stdout, *) '  reactions= ', pr_reaction_num
  write(stdout, *) '  plasma-material_interactions= ', pr_pmi_num
  write(stdout, *) 'Test_species_are:'
  do i = 1, pr_test_num
    write(stdout, *) pr_test(i), ', ', trim(sp_name(pr_test(i)))
  end do
  write(stdout, *) 'Background_species_are:'
  do i = 1, pr_background_num
    write(stdout, *) pr_background(i), ', ', trim(sp_name(pr_background(i)))
  end do
  write(stdout, *) 'Reference_material_indices_are:'
  do i = 1, pr_materials_num
    write(stdout, *) pr_mat_ref(i), ', ', trim(ma_name(pr_mat_ref(i)))
  end do
  write(stdout, *) 'Reference_reaction_indices_are:'
  if (pr_reaction_num > 0) then
    do i = 1, pr_reaction_num
      write(stdout, *) pr_reaction(i), ', ', trim(rc_name(pr_reaction(i)))
    end do
  end if

```

```

end if
write(stdout, *) 'Reference_PMI_indices_are:'
do i = 1, pr_pmi_num
  write(stdout, *) pr_pm_ref(i), ', ', trim(pm_name(pr_pm_ref(i)))
end do

var_alloc(problem_reaction_num)
var_alloc(problem_test_reaction)
var_alloc(problem_test_background)
var_alloc(problem_pmi_case_num)
var_alloc(problem_pmi_cases)
var_alloc(problem_pmi_num_arrange)
var_alloc(problem_pmi_prod_mult)
var_alloc(problem_pmi_products)
var_alloc(problem_num_arrangements)
var_alloc(problem_prod_mult)
var_alloc(problem_test_products)

assert(pr_max_equiv ≥ max(pr_test_num, pr_background_num))
var_alloc(num_equiv)
var_alloc(equivalents)

do i = 1, pr_test_num
  do j = 1, pr_reaction_max
    pr_ts_rc(i, j) = int_unused
    pr_ts_bk(i, j) = int_unused
    pr_num_arrangements(i, j) = int_unused
  do k = 1, pr_arrangement_max
    pr_prod_mult(i, j, k) = real_unused
    do l = 1, rc_product_max
      pr_ts_prod(i, j, k, l) = int_unused
    end do
  end do
end do
end do

do i = 1, pr_max_equiv
  do j = 1, num_generics
    num_equiv_j = 0
    equivalents_i_j = 0
  end do
end do

/* For each test species, first obtain the index of the corresponding generic archetype, generic,
   increment the number of equivalents to this archetype, and store the species index of this test in
   the equivalents array. */
do i = 1, pr_test_num
  generic = generics_pr_test(i)
  num_equiv_generic ++
  equivalents_num_equiv_generic, generic = pr_test(i)
end do

/* Do same for generics of background species. Since the same species may be a test, as well as a
   background, check to see if it's already on the list (call to int_lookup). */
do i = 1, pr_background_num

```

```

generic = genericspr_background(i)
if ((num_equivgeneric  $\equiv$  0)  $\vee$  (int_lookup(pr_background(i), equivalents1, generic, pr_max_equiv)  $\equiv$  0))
  then
    num_equivgeneric ++
    equivalentsnum_equivgeneric, generic = pr_background(i)
  endif
end do

/* We can only deal with binary collisions currently */
if (pr_reaction_num > 0) then
  do j = 1, pr_reaction_num
    assert(rc_reagent_num(pr_reaction(j))  $\equiv$  2)
  end do
end if

do i = 1, pr_test_num
  pr_rc_num(i) = 0
end do
pr_bkrc_dim = 0

if (pr_reaction_num > 0) then
  do j = 1, pr_reaction_num

    /* The original version of this code permitted an arbitrary ordering of test and background
       in reagent specification (restricted to binary collisions). With the introduction of
       neutral-neutral elastic scattering reactions (reagents having the same species), a specific
       ordering had to be chosen. The convention used in many outside data sources (e.g.,
       Aladdin) is consistent with “background + test”. So, this ordering is chosen here as well.
       Note: assuming that we do not need code here to deal with species-specific reactions
       (asserts to follow should catch exceptions). */

    ts_reagent = 0
    do i = 1, pr_bkrc_reagent_max
      generic_reagentsi = 0
    end do
    do i = 1, rc_reagent_num(pr_reaction(j))
      generic_reagentsi = genericsrc_reagent(pr_reaction(j), i)
      if (pr_test_lookup(equivalents1, generic_reagentsi)  $\neq$  0) then
        ts_reagent = 2    // else: is a background reaction (below)
        bk_reagent = 1
      end if
    end do

    /* Done setting generic_reagents; use to assign remaining pr arrays */
    if (ts_reagent > 0) then
      if (rc_gen(pr_reaction(j))  $\equiv$  rc_generic_no) then
        test = pr_test_lookup(rc_reagent(pr_reaction(j), ts_reagent))
        background = pr_background_lookup(rc_reagent(pr_reaction(j), bk_reagent))
        assert(pr_test_check(test)  $\wedge$  pr_background_check(background))
        pr_rc_num(test) ++
        assert(pr_rc_num(test)  $\leq$  pr_reaction_max)
        pr_ts_rc(test, pr_rc_num(test)) = j
        pr_ts_bk(test, pr_rc_num(test)) = background
        pr_num_arrangements(test, pr_rc_num(test)) = 1
        pr_prod_mult(test, pr_rc_num(test), 1) = one
        do k = 1, rc_product_num(pr_reaction(j))

```

```

    pr_ts_prod(test, pr_rc_num(test), 1, k) = rc_product(pr_reaction(j), k)
  end do
else if (rc_gen(pr_reaction(j))  $\equiv$  rc_generic_yes) then
  do k = 1, num_equiv_generic_reagents_ts_reagent
    test = pr_test_lookup(equivalents_k, generic_reagents_ts_reagent)
    do l = 1, num_equiv_generic_reagents_bk_reagent
      background = pr_background_lookup(equivalents_l, generic_reagents_bk_reagent)
      assert(pr_test_check(test)  $\wedge$  pr_background_check(background))
      pr_rc_num(test)++
      assert(pr_rc_num(test)  $\leq$  pr_reaction_max)
      pr_ts_rc(test, pr_rc_num(test)) = j
      pr_ts_bk(test, pr_rc_num(test)) = background
      reagents_bk_reagent = pr_background(background)
      reagents_ts_reagent = pr_test(test)
      call set_products(pr_test_args(test), sp_args(reagents), rc_args(pr_reaction(j)))
    end do
  end do
else
  assert('Illegal_value_of_rc_gen(j)'  $\equiv$  '␣')
end if
else /* Reactions among background species... We have not established a protocol yet for
dealing with reactions involving two background species. The typical reactions involve a
test species (given velocity and position) and a background species (described only by a
temperature, density, and flow velocity). If we're going to treat reactions between two
background species in the same way, we'll need to specify which behaves more like a test
particle. Rather than rely on a convention for the ordering of the reagents on input, make
this decision here based on mass. The obvious motivation is that one of the reagents is
inevitably an electron in which case i) its temperature determines the reaction rate, and
ii) we will not likely be interested in it as a test particle. Whereas, the other reagent
(e.g., an ion about to be recombined) will be sampled in preparation for determining the
velocity of the sourced test species. */
if (rc_gen(pr_reaction(j))  $\equiv$  rc_generic_no) then
  do k = 1, pr_bkrc_reagent_max
    back_k = pr_background_lookup(rc_reagent(pr_reaction(j), k))
    assert(pr_background_check(back_k))
  end do
  pr_bkrc_dim++
  var_realloca(problem_background_reaction)
  var_realloca(problem_bkrc_reagents)
  var_realloca(problem_bkrc_products)
  pr_bkrc(pr_bkrc_dim) = j
  if (sp_m(pr_background(back_2)) < sp_m(pr_background(back_1))) then
    pr_bkrc_rg(pr_bkrc_dim, 1) = back_2
    pr_bkrc_rg(pr_bkrc_dim, 2) = back_1
  else
    pr_bkrc_rg(pr_bkrc_dim, 1) = back_1
    pr_bkrc_rg(pr_bkrc_dim, 2) = back_2
  end if
  assert(rc_product_num(pr_reaction(j))  $\equiv$  1)
  // Otherwise, we need to add machinery to deal with arrangements, etc.
  pr_bkrc_prod(pr_bkrc_dim, 1) = rc_product(pr_reaction(j), 1)

```

```

else if (rc_gen(pr_reaction(j)) ≡ rc_generic_yes) then
  do k = 1, num_equiv_generic_reagents_1
    back_1 = pr_background_lookup(equivalents_k, generic_reagents_1)
    assert(pr_background_check(back_1))
  do l = 1, num_equiv_generic_reagents_pr_bkrc_reagent_max
    back_pr_bkrc_reagent_max =
      pr_background_lookup(equivalents_l, generic_reagents_pr_bkrc_reagent_max)
    assert(pr_background_check(back_pr_bkrc_reagent_max))
    pr_bkrc_dim++
    var_realloca(problem_background_reaction)
    var_realloca(problem_bkrc_reagents)
    var_realloca(problem_bkrc_products)
    pr_bk_rc(pr_bkrc_dim) = j
  do m = 1, pr_bkrc_reagent_max
    if (sp_m(pr_background(back_pr_bkrc_reagent_max)) < sp_m(pr_background(back_1)))
      then
        pr_bkrc_rg(pr_bkrc_dim, m) = back_pr_bkrc_reagent_max - m + 1
      else
        pr_bkrc_rg(pr_bkrc_dim, m) = back_m
      end if
      reagents_m = pr_background(back_m)
    end do
    call set_bkrc_products(pr_bkrc_dim, sp_args(reagents))
  end do
end do
else
  assert('Illegal_value_of_rc_gen(j)' ≡ ' ')
end if
end if
end do
end if /* Have to introduce pr_bkrc_dim because netCDF allocates space for these arrays even if
pr_bkrc_num=0. The problem arises when the netCDF file containing these variables is read back
in. The size is reported to be non-zero, but the dimensioning variables says it should be zero.
The awkward solution is to use pr_bkrc_dim as the dimension of these arrays and pr_bkrc_num as
the actual number of reactions. The former is set to 1 if there are no background reactions. */
pr_bkrc_num = pr_bkrc_dim
if (pr_bkrc_dim ≡ 0) then
  pr_bkrc_dim = 1
  var_alloc(problem_background_reaction)
  var_alloc(problem_bkrc_reagents)
  var_alloc(problem_bkrc_products)
  pr_bk_rc(1) = int_unused
  do i = 1, pr_bkrc_reagent_max
    pr_bkrc_rg(1, i) = int_unused
  end do
  do i = 1, rc_product_max
    pr_bkrc_prod(1, i) = int_unused
  end do
else
  var_reallocb(problem_background_reaction)
  var_reallocb(problem_bkrc_reagents)

```

```

    var_reallocb(problem_bkrc_products)
end if

do i = 1, pr_test_num
    pr_pm_case_num(i) = 0
    do j = 1, pr_pmi_max
        pr_pm_cases(i, j) = int_unused
        pr_pm_num_arrange(i, j) = int_unused
        do k = 1, pr_arrangement_max
            pr_pm_prod_mult(i, j, k) = real_unused
            do l = 1, rc_product_max
                pr_pm_prod(i, j, k, l) = int_unused
            end do
        end do
    end do
end do

end do
/* Here is the analogous code for PMI which sets up the list of PMI in which each test
species participates. Species-specific PMI are dealt with first: */
do j = 1, pr_pmi_num
    if (pm_gen(pr_pm_ref(j)) ≡ pmi_generic_no) then
        test = pr_test_lookup(pm_reagent(pr_pm_ref(j)))
        pr_pm_case_num(test)++
        pr_pm_cases(test, pr_pm_case_num(test)) = j

        /* For species-specific PMI, the products can be set directly: */
        pr_pm_num_arrange(test, pr_pm_case_num(test)) = 1
        pr_pm_prod_mult(test, pr_pm_case_num(test), 1) = one
        do k = 1, pm_product_num(pr_pm_ref(j))
            pr_pm_prod(test, pr_pm_case_num(test), 1, k) = pm_product(pr_pm_ref(j), k)
        end do

        /* For PMI which refer to generic reagents and product species, proceed as above for
reactions. The process is simplified because (1) there are no generic materials, (2) the
ordering of test and materials is hardwired. */
    else if (pm_gen(pr_pm_ref(j)) ≡ pmi_generic_yes) then
        generic_reagents_1 = generics_pm_reagent(pr_pm_ref(j))
        do k = 1, num_equiv_generic_reagents_1
            test = pr_test_lookup(equivalents_k, generic_reagents_1)
            assert(pr_test_check(test))
            pr_pm_case_num(test)++
            pr_pm_cases(test, pr_pm_case_num(test)) = j

            call set_pmi_products(pr_test_args(test), pm_args(pr_pm_ref(j)))
        end do
    else
        assert('Illegal_value_of_pm_gen(j)' ≡ ' ')
    end if
end do

return
end

```

This is a clearing-house routine used to select the product-setting routine specific for each reaction.

⟨ Functions and Subroutines 1.2 ⟩ +=

```

subroutine set_products(pr_test_dummy(ts), sp_dummy(reagents), rc_dummy(r))
  implicit none f77
  pr_common
  rc_common
  implicit none f90

  pr_test_decl(ts)    // Input
  rc_decl(r)
  sp_decl(reagents*)

  integer i    // Local
  real sum

  if (rc_reaction_type(r) ≡ 'ionize') then
    call set_prod_ionize(pr_test_args(ts), sp_args(reagents), rc_args(r))
  else if (rc_reaction_type(r) ≡ 'ionize_suppress') then
    call set_prod_ionize(pr_test_args(ts), sp_args(reagents), rc_args(r))
  else if (rc_reaction_type(r) ≡ 'chargex') then
    call set_prod_chargex(pr_test_args(ts), sp_args(reagents), rc_args(r))
  else if (rc_reaction_type(r) ≡ 'dissoc') then
    call set_prod_dissoc(pr_test_args(ts), sp_args(reagents), rc_args(r))
  else if (rc_reaction_type(r) ≡ 'dissoc_rec') then
    call set_prod_dissoc_rec(pr_test_args(ts), sp_args(reagents), rc_args(r))
  else if (rc_reaction_type(r) ≡ 'dissoc_cramd') then
    call set_prod_dissoc(pr_test_args(ts), sp_args(reagents), rc_args(r))
  else if (rc_reaction_type(r) ≡ 'elastic') then
    call set_prod_elastic(pr_test_args(ts), sp_args(reagents), rc_args(r))
  else if (rc_reaction_type(r) ≡ 'bgk_elastic') then
    call set_prod_elastic(pr_test_args(ts), sp_args(reagents), rc_args(r))
  else if (rc_reaction_type(r) ≡ 'excitation') then
    call set_prod_excite(pr_test_args(ts), sp_args(reagents), rc_args(r))
  else if (rc_reaction_type(r) ≡ 'deexcitation') then
    call set_prod_excite(pr_test_args(ts), sp_args(reagents), rc_args(r))
  else
    pr_num_arrangements(ts, pr_rc_num(ts)) = 0
    return
  end if

  /* Rescale multiplicity so that it becomes a probability */
  sum = zero
  do i = 1, pr_num_arrangements(ts, pr_rc_num(ts))
    sum = sum + pr_prod_mult(ts, pr_rc_num(ts), i)
  end do
  do i = 1, pr_num_arrangements(ts, pr_rc_num(ts))
    pr_prod_mult(ts, pr_rc_num(ts), i) = pr_prod_mult(ts, pr_rc_num(ts), i) / sum
  end do

  return
end

```

Set up isotopic variants of ionizations. By convention, an ionization reaction specification takes the form:



note that B could be charged to start with.

⟨ Functions and Subroutines 1.2 ⟩ +≡

```

subroutine set_prod_ionize(pr_test_dummy(ts), sp_dummy(reagents), rc_dummy(r))
  implicit_none_f77
  ps_common
  rc_common
  pr_common
  sp_common
  implicit_none_f90

  pr_test_decl(ts)
  sp_decl(reagents*)
  rc_decl(r)

  external species_add_check    // External
  logical species_add_check

  integer i    // Local

  sp_decl(product2)

  st_decls

  assert(rc_product_num(r) ≡ 3)
  pr_num_arrangements(ts, pr_rc_num(ts)) = 1
  pr_prod_mult(ts, pr_rc_num(ts), 1) = one

  /* First product corresponds to the reagent which does the ionizing: its species is unaltered. */
  assert(sp_generic(rc_product(r, 1)) ≡ sp_generic(reagents1))
  pr_ts_prod(ts, pr_rc_num(ts), 1, 1) = reagents1

  /* By convention, the third (last) product is the electron knocked off during ionization. */
  assert(rc_product(r, 3) ≡ sp_lookup('e'))
  pr_ts_prod(ts, pr_rc_num(ts), 1, 3) = rc_product(r, 3)

  /* The second product is the result of the ionization. It corresponds to the second reagent, minus
     an electron. By looping over the generic equivalents to the second rc_product, we ensure that any
     details (e.g., quantum state) not checked by species_add_check are correctly carried through. */
  product2 = rc_product(r, 3)
  do i = 1, num_equivgenerics rc_product(r, 2)
    product1 = equivalentsi, generics rc_product(r, 2)
    if (species_add_check(2, sp_args(product), 1, sp_args(reagents2))) then
      pr_ts_prod(ts, pr_rc_num(ts), 1, 2) = product1
    return
  end if
end do
  assert('Unmatched_ionization_product' ≡ '␣')

  return
end

```

Set up isotopic variants of charge exchange. By convention, a charge exchange reaction specification takes the form:



the superscript “+”s are used only to indicate a change in charge state; the actual charge states of A and B are intended to be arbitrary. The procedure below can handle both single and double electron charge exchange.

⟨Functions and Subroutines 1.2⟩ +=

```
subroutine set_prod_chargex(pr_test_dummy(ts), sp_dummy(reagents), rc_dummy(r))
  implicit_none_f77
  rc_common
  pr_common
  sp_common
  ps_common
  implicit_none_f90

  pr_test_decl(ts)
  rc_decl(r)
  sp_decl(reagents_*)

  external species_add_check    // External
  logical species_add_check

  integer i, num_electrons    // Local

  sp_decl(product_3)

  st_decls

  num_electrons = 0
  assert(rc_product_num(r) ≡ 2)
  pr_num_arrangements(ts, pr_rc_num(ts)) = 1
  pr_prod_mult(ts, pr_rc_num(ts), 1) = one
  /* The first product should match up with the first reagent plus one or two electrons. */
  product_1 = sp_lookup('e')
  product_2 = reagents_1
  do i = 1, num_equiv_generics_rc_product(r, 1)
    if (species_add_check(2, sp_args(product), 1, sp_args(equivalents_i, generics_rc_product(r, 1)))) then
      pr_ts_prod(ts, pr_rc_num(ts), 1, 1) = equivalents_i, generics_rc_product(r, 1)
      num_electrons = 1
      goto loop1
    else    // Might be double charge exchange
      product_3 = product_1
      if (species_add_check(3, sp_args(product), 1, sp_args(equivalents_i, generics_rc_product(r, 1)))) then
        pr_ts_prod(ts, pr_rc_num(ts), 1, 1) = equivalents_i, generics_rc_product(r, 1)
        num_electrons = 2
        goto loop1
      end if
    end if
  end do
  assert('Unmatched_first_charge_exchange_product' ≡ ' ')

loop1: continue
  /* Likewise, the second reagent should match the second product plus one or two electrons. */
  do i = 1, num_equiv_generics_rc_product(r, 2)
```

```

    product2 = equivalentsi, genericsrc-product(r, 2)
if ((num_electrons ≡ 1) ∧ (species_add_check(2, sp_args(product), 1, sp_args(reagents2)))) then
    pr_ts_prod(ts, pr_rc_num(ts), 1, 2) = product2
    return
else if (num_electrons ≡ 2) then
    if (species_add_check(3, sp_args(product), 1, sp_args(reagents2))) then
        pr_ts_prod(ts, pr_rc_num(ts), 1, 2) = product2
        return
    end if
end if
end do
assert('Unmatched_second_charge_exchange_product' ≡ '␣')
return
end

```

Set up isotopic variants of elastic collisions. By convention, an elastic reaction specification takes the form:



the superscript “+”s are used only to indicate a charge state; the actual charge states of A and B are intended to be arbitrary. Currently the only reaction allowed is that given below.



⟨ Functions and Subroutines 1.2 ⟩ +≡

```

subroutine set_prod_elastic(pr_test_dummy(ts), sp_dummy(reagents), rc_dummy(r))
  implicit none
  rc_common
  pr_common
  sp_common
  implicit none

  pr_test_decl(ts)
  rc_decl(r)
  sp_decl(reagents*)

  external species_add_check    // External
  logical species_add_check

  st_decls

  assert(rc_product_num(r) ≡ 2)
  pr_num_arrangements(ts, pr_rc_num(ts)) = 1
  pr_prod_mult(ts, pr_rc_num(ts), 1) = one

  assert(sp_generic(rc_product(r, 1)) ≡ sp_generic(reagents1))
@#if 0
  /* Having tested reagents and stated assumptions in the introduction, why did we write this at
  all? */
  pr_ts_prod(ts, pr_rc_num(ts), 1, 1) = rc_product(r, 1)
@#endif
  pr_ts_prod(ts, pr_rc_num(ts), 1, 1) = reagents1
  assert(sp_generic(rc_product(r, 2)) ≡ sp_generic(reagents2))
@#if 0
  pr_ts_prod(ts, pr_rc_num(ts), 1, 2) = rc_product(r, 2)
@#endif
  pr_ts_prod(ts, pr_rc_num(ts), 1, 2) = reagents2

  return
end

```

Set up isotopic variants of dissociation. This reactions is assumed to have the form:



where the products B_j are the result of the dissociation of B.

⟨ Functions and Subroutines 1.2 ⟩ +=

```

subroutine set_prod_dissoc(pr_test_dummy(ts), sp_dummy(reagents), rc_dummy(r))
  implicit none_f77
  rc_common
  pr_common
  sp_common
  implicit none_f90

  pr_test_decl(ts)
  rc_decl(r)
  sp_decl(reagents*)

  integer i     // Local

  assert(sp_generic(rc_product(r, 1)) ≡ sp_generic(reagents1))

  if (rc_product_num(r) ≡ 3) then
    call product_perm_2(pr_test_args(ts), 1, sp_args(reagents2), rc_args(r))
  else if (rc_product_num(r) ≡ 4) then
    call product_perm_3(pr_test_args(ts), 1, sp_args(reagents2), rc_args(r))
  else
    assert('Illegal_value_of_rc_product_num' ≡ '_')
  end if

  do i = 1, pr_num_arrangements(ts, pr_rc_num(ts))
    pr_ts_prod(ts, pr_rc_num(ts), i, 1) = reagents1
  end do

  return
end

```

Set up isotopic variants of dissociative recombination. This reactions is assumed to have the form:



where the products C_j are the result of the dissociation of C which is formed by the recombination of A and B.

⟨ Functions and Subroutines 1.2 ⟩ +=

```
subroutine set_prod_dissoc_rec(pr_test_dummy(ts), sp_dummy(reagents), rc_dummy(r))
  implicit_none_f77
  rc_common
  pr_common
  sp_common
  implicit_none_f90

  pr_test_decl(ts)
  rc_decl(r)
  sp_decl(reagents*)

  if (rc_product_num(r) ≡ 2) then
    call product_perm_2(pr_test_args(ts), 2, sp_args(reagents), rc_args(r))
  else if (rc_product_num(r) ≡ 3) then
    call product_perm_3(pr_test_args(ts), 2, sp_args(reagents), rc_args(r))
  else
    assert('Illegal_value_of_rc_product_num' ≡ ' ')
  end if

  return
end
```

Set up isotopic variants of excitation. This routine is also suitable for deexcitation. The general form of these reactions is:



where A is the background species and B is the test species. The * indicates a change in the electron configuration of the species. In terms of DEGAS 2's species characteristics, this is just a change in the species label; the elements comprising the reagent and product test species must be the same.

(Functions and Subroutines 1.2) +=

```
subroutine set_prod_excite(pr_test_dummy(ts), sp_dummy(reagents), rc_dummy(r))
  implicit_none_f77
  ps_common
  rc_common
  pr_common
  sp_common
  implicit_none_f90

  pr_test_decl(ts)
  sp_decl(reagents_*)
  rc_decl(r)

  external species_add_check    // External
  logical species_add_check

  integer i    // Local

  sp_decl(product)

  st_decls

  assert(rc_product_num(r) == 2)
  pr_num_arrangements(ts, pr_rc_num(ts)) = 1
  pr_prod_mult(ts, pr_rc_num(ts), 1) = one
  /* First product corresponds to the reagent which does the exciting: its species is unaltered. */
  assert(sp_generic(rc_product(r, 1)) == sp_generic(reagents_1))
  pr_ts_prod(ts, pr_rc_num(ts), 1, 1) = reagents_1 /* The second product is the result of the
  excitation or deexcitation. It corresponds only to a different electron configuration from the
  reagent. So, species_add_check will suffice for picking out the appropriate product. */
  do i = 1, num_equiv_generics_rc_product(r, 2)
    product = equivalents_i_generics_rc_product(r, 2)
    if (species_add_check(1, sp_args(product), 1, sp_args(reagents_2))) then
      pr_ts_prod(ts, pr_rc_num(ts), 1, 2) = product
      return
    end if
  end do
  assert('Unmatched_excitation_product' == '_')

  return
end
```

This is a clearing-house routine used to select the product-setting routine specific for each reaction among background species. Presently, the only application for this is recombination.

⟨ Functions and Subroutines 1.2 ⟩ +≡

```

subroutine set_bkrc_products(bkrc_ind, sp_dummy(reagents))
  implicit none f77
  pr_common
  rc_common
  implicit none f90

  integer bkrc_ind    // Input

  sp_decl(reagents*)
  integer i    // Local

  do i = 1, rc_product_max
    pr_bkrc_prod(bkrc_ind, i) = int_unused
  end do
  if (rc_reaction_type(pr_reaction(pr_bk_rc(bkrc_ind))) ≡ 'recombination') then
    call set_prod_recombine(bkrc_ind, sp_args(reagents))
  else
    assert('Unsupported_reaction_type' ≡ '␣')
    return
  end if

  return
end

```

Set up isotopic variants of recombinations. By convention, a recombination reaction specification takes the form:



note that B could be charged to start with.

⟨ Functions and Subroutines 1.2 ⟩ +≡

```
subroutine set_prod_recombine(bkrc_ind, sp_dummy(reagents))
  implicit_none_f77
  rc_common
  pr_common
  sp_common
  ps_common
  implicit_none_f90

  integer bkrc_ind
  sp_decl(reagents_*)

  external species_add_check    // External
  logical species_add_check

  integer i    // Local
  rc_decl(r)
  sp_decl(product_2)

  st_decls

  r = pr_reaction(pr_bk_rc(bkrc_ind))
  assert(rc_product_num(r) ≡ 1)

  /* Assuming that only one product is listed for recombination. If there are more products, not
   only does this routine have to be augmented, but will probably need to add common arrays
   to deal with product permutations. By looping over the generic equivalents to rc_product, we
   ensure that any details (e.g., quantum state) not checked by species_add_check are correctly
   carried through. */
  do i = 1, num_equiv_generics_rc_product(r, 1)
    product_1 = equivalents_i, generics_rc_product(r, 1)
    if (species_add_check(1, sp_args(product), 2, sp_args(reagents))) then
      pr_bkrc_prod(bkrc_ind, 1) = product_1
      return
    end if
  end do
  assert('Unmatched_recombination_product' ≡ '␣')

  return
end
```

This is a clearing-house routine used to select the product-setting routine specific for each PMI.

⟨ Functions and Subroutines 1.2 ⟩ +=

```

subroutine set_pmi_products(pr_test_dummy(ts), pm_dummy(p))
  implicit none_f77
  pr_common
  pm_common
  implicit none_f90

  pr_test_decl(ts)    // Input
  pm_decl(p)

  integer i    // Local
  real sum

  if (pm_pmi_type(p) ≡ 'reflection') then
    call set_prod_reflection(pr_test_args(ts), pm_args(p))
  else if (pm_pmi_type(p) ≡ 'adsorption') then
    call set_prod_adsorption(pr_test_args(ts), pm_args(p))
  else if (pm_pmi_type(p) ≡ 'desorption') then
    call set_prod_desorption(pr_test_args(ts), pm_args(p))
  else if (pm_pmi_type(p) ≡ 'sputter') then
    assert('set_product_for_sputter_not_available' ≡ '␣')
  end if

  /* Rescale multiplicity so that it becomes a probability */
  sum = zero
  do i = 1, pr_pm_num_arrange(ts, pr_pm_case_num(ts))
    sum = sum + pr_pm_prod_mult(ts, pr_pm_case_num(ts), i)
  end do
  if (sum > zero) then    // In some cases, pr_pm_prod_mult is ≡ 0.
    do i = 1, pr_pm_num_arrange(ts, pr_pm_case_num(ts))
      pr_pm_prod_mult(ts, pr_pm_case_num(ts), i) = pr_pm_prod_mult(ts, pr_pm_case_num(ts), i) / sum
    end do
  end if

  return
end

```

Set up isotopic variations of reflections. This is a simple case of “what comes in must go out.”

⟨ Functions and Subroutines 1.2 ⟩ +≡

```
subroutine set_prod_reflection(pr_test_dummy(ts), pm_dummy(p))
  implicit none f77
  pr_common      // Common
  pm_common
  sp_common
  implicit none f90

  pr_test_decl(ts)    // Input
  pm_decl(p)

  assert(pm_product_num(p) ≡ 1)
  pr_pm_num_arrange(ts, pr_pm_case_num(ts)) = 1
  pr_pm_prod_mult(ts, pr_pm_case_num(ts), 1) = one

  /* The only product had better be the same as the incoming test particle */
  assert(sp_generic(pm_product(p, 1)) ≡ sp_generic(pr_test(ts)))
  pr_pm_prod(ts, pr_pm_case_num(ts), 1, 1) = pr_test(ts)

  return
end
```

Set up isotopic variations for adsorptions. It doesn't get any easier than this.

⟨ Functions and Subroutines 1.2 ⟩ +≡

```
subroutine set_prod_adsorption(pr_test_dummy(ts), pm_dummy(p))
  implicit none f77
  pr_common      // Common
  pm_common
  implicit none f90

  pr_test_decl(ts)    // Input
  pm_decl(p)

  assert(pm_product_num(p) ≡ 0)
  pr_pm_num_arrange(ts, pr_pm_case_num(ts)) = 1
  pr_pm_prod_mult(ts, pr_pm_case_num(ts), 1) = one

  pr_pm_prod(ts, pr_pm_case_num(ts), 1, 1) = 0

  return
end
```

Set up isotopic variations for desorptions. The assumptions made here reflect the algorithm used in the original DEGAS code. Namely, for most incident neutral species, the desorbed species is the same as the incident one. For an incident hydrogen atom, however, the possibility existed for desorption as a molecule. It is intended that this option be represented by two products in the reference description of the PMI, only one of which is used at a time here. In the event that a molecule is returned, the isotope of one of the atoms is required to match that of the incident H. The other atom will be assigned in a general way here based on the other isotopes present in the problem. It is assumed that the weights of these various options are established in the particular data file for this PMI and will be implemented only at run time. An analogous description holds for carbon as well as the various methane fragments.

⟨Functions and Subroutines 1.2⟩ +≡

```

subroutine set_prod_desorption(pr_test_dummy(ts), pm_dummy(p))
  implicit none f77
  pr_common      // Common
  pm_common
  el_common
  ps_common
  implicit none f90

  pr_test_decl(ts)  // Input
  pm_decl(p)

  external species_el_count    // External
  integer species_el_count

  integer i, ic, j, count_ts, count_prod    // Local
  logical match

  pr_generic_decl(generic_prod)

  pr_pm_num_arrange(ts, pr_pm_case_num(ts)) = 0
  do i = 1, pm_product_num(p)
    generic_prod = generics_pm_product(p, i)
    assert(generic_prod > 0 ∧ generic_prod ≤ num_generics)
    do j = 1, num_equiv_generic_prod
      match =  $\mathcal{T}$ 
      do ic = 1, el_num
        count_ts = species_el_count(pr_test(ts), el_args(ic))
        count_prod = species_el_count(equivalents_j_generic_prod, el_args(ic))
        if (count_ts > count_prod)
          match =  $\mathcal{F}$ 
        end do
      end do
      if (match) then
        pr_pm_num_arrange(ts, pr_pm_case_num(ts))++
        /* Set this to zero; it will be determined at run time. */
        pr_pm_prod_mult(ts, pr_pm_case_num(ts), pr_pm_num_arrange(ts,
          pr_pm_case_num(ts))) = zero
        pr_pm_prod(ts, pr_pm_case_num(ts), pr_pm_num_arrange(ts, pr_pm_case_num(ts)),
          1) = equivalents_j_generic_prod
      end if
    end do
  end do
  assert(pr_pm_num_arrange(ts, pr_pm_case_num(ts)) > 0)

  return
end

```

Assemble and check permutations of two products arising from the dissociation of *num_sp* species *sp* in reaction *r*. The test species *ts* is carried into the routine since to be used as an array index. Note that for simplicity duplicates (i.e., permuted lists of equivalent species) are not eliminated. This makes for more arrangements than necessary, but the overall probability for a given unique arrangement is effected nonetheless. If desired, this duplication could be eliminated through additional logic.

⟨ Functions and Subroutines 1.2 ⟩ +=

```
subroutine product_perm_2(pr_test_dummy(ts), num_sp, sp_dummy(sp), rc_dummy(r))
  implicit none f77
  rc_common
  sp_common
  pr_common
  ps_common
  implicit none f90

  integer num_sp    // Input

  pr_test_decl(ts)
  sp_decl(spnum_sp)
  rc_decl(r)

  external species_add_check    // External
  logical species_add_check

  integer i, j, first_product    // Local

  sp_decl(product2)
  pr_generic_decl(generic1)
  pr_generic_decl(generic2)

  /* A num_sp of 1 indicates pure dissociation ⇒ the actual first product will be the dissociating
     agent; the first dissociation product is then the second product. On the other hand, num_sp =
     2 suggests dissociative recombination where all products come from dissociation. */
  if (num_sp ≡ 1) then
    first_product = 2
  else if (num_sp ≡ 2) then
    first_product = 1
  else
    assert('Invalid_num_sp' ≡ '␣')
  end if

  /* Obtain generic indices of the two products */
  generic_1 = genericsrc_product(r, first_product)
  generic_2 = genericsrc_product(r, first_product+1)

  /* Loop over the species equivalent to each generic and see which combinations are consistent
     with the given reagents. */
  pr_num_arrangements(ts, pr_rc_num(ts)) = 0
  do i = 1, num_equivgeneric_1
    product1 = equivalentsi, generic_1
    do j = 1, num_equivgeneric_2
      product2 = equivalentsj, generic_2
      if (species_add_check(2, sp_args(product), num_sp, sp_args(sp))) then
        pr_num_arrangements(ts, pr_rc_num(ts))++    // Valid arrangement
        assert(pr_num_arrangements(ts, pr_rc_num(ts)) ≤ pr_arrangement_max)
        pr_ts_prod(ts, pr_rc_num(ts), pr_num_arrangements(ts, pr_rc_num(ts)),
                   first_product) = product1
```

```
    pr_ts_prod(ts, pr_rc_num(ts), pr_num_arrangements(ts, pr_rc_num(ts)),  
              first_product + 1) = product2  
    pr_prod_mult(ts, pr_rc_num(ts), pr_num_arrangements(ts,  
                  pr_rc_num(ts))) = sp_multiplicity(product1) * sp_multiplicity(product2)  
  end if  
end do  
end do  
assert(pr_num_arrangements(ts, pr_rc_num(ts)) > 0)  
return  
end
```

Assemble and check permutations of three products arising from the dissociation of species *sp* in reaction *r*. The test species *ts* is carried into the routine since to be used as an array index. Note that for simplicity duplicates (i.e., permuted lists of equivalent species) are not eliminated. This makes for more arrangements than necessary, but the overall probability for a given unique arrangement is effected nonetheless. If desired, this duplication could be eliminated through additional logic.

⟨Functions and Subroutines 1.2⟩ +≡

```
subroutine product_perm_3(pr_test_dummy(ts), num_sp, sp_dummy(sp), rc_dummy(r))
  implicit none f77
  rc_common
  sp_common
  pr_common
  ps_common
  implicit none f90

  integer num_sp     // Input

  pr_test_decl(ts)
  sp_decl(spnum_sp)
  rc_decl(r)

  external species_add_check     // External
  logical species_add_check

  integer i, j, k, first_product     // Local

  sp_decl(product3)
  pr_generic_decl(generic_1)
  pr_generic_decl(generic_2)
  pr_generic_decl(generic_3)

  /* A num_sp of 1 indicates pure dissociation ⇒ the actual first product will be the dissociating
     agent; the first dissociation product is then the second product. On the other hand, num_sp =
     2 suggests dissociative recombination where all products come from dissociation. */
  if (num_sp ≡ 1) then
    first_product = 2
  else if (num_sp ≡ 2) then
    first_product = 1
  else
    assert('Invalid_num_sp' ≡ '␣')
  end if

  /* Obtain generic indices for each product */
  generic_1 = genericsrc_product(r, first_product)
  generic_2 = genericsrc_product(r, first_product+1)
  generic_3 = genericsrc_product(r, first_product+2)

  /* Loop over the species equivalent to each generic and see which combinations are consistent
     with the given reagents. */
  pr_num_arrangements(ts, pr_rc_num(ts)) = 0
  do i = 1, num_equivgeneric_1
    product1 = equivalentsi, generic_1
    do j = 1, num_equivgeneric_2
      product2 = equivalentsj, generic_2
      do k = 1, num_equivgeneric_3
        product3 = equivalentsk, generic_3
        if (species_add_check(3, sp_args(product), num_sp, sp_args(sp))) then
```

```

    pr_num_arrangements(ts, pr_rc_num(ts))++    // Valid arrangement
    assert(pr_num_arrangements(ts, pr_rc_num(ts)) ≤ pr_arrangement_max)
    pr_ts_prod(ts, pr_rc_num(ts), pr_num_arrangements(ts, pr_rc_num(ts)),
        first_product) = product1
    pr_ts_prod(ts, pr_rc_num(ts), pr_num_arrangements(ts, pr_rc_num(ts)),
        first_product + 1) = product2
    pr_ts_prod(ts, pr_rc_num(ts), pr_num_arrangements(ts, pr_rc_num(ts)),
        first_product + 2) = product3
    pr_prod_mult(ts, pr_rc_num(ts), pr_num_arrangements(ts, pr_rc_num(ts))) =
        sp_multiplicity(product1) * sp_multiplicity(product2) * sp_multiplicity(product3)
    end if
  end do
end do
end do
assert(pr_num_arrangements(ts, pr_rc_num(ts)) > 0)
return
end

```

2 References

3 INDEX

assert: 1.2, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 1.11, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.19, 1.20, 1.21, 1.22, 1.23, 1.24, 1.25, 1.26.

b: [1.4](#), [1.5](#), [1.6](#), [1.7](#), [1.8](#).

back: 1.11.

background: 1.11.

bk_reagent: [1.11](#).

bkr_ind: [1.19](#), [1.20](#).

char_undef: 1.2.

char_uninit: 1.9.

check_problem: 1.2, [1.11](#).

count_prod: [1.24](#).

count_ts: [1.24](#).

datasetup: 1.

diskin: 1.2.

e: [1.4](#), [1.5](#), [1.6](#), [1.7](#), [1.8](#).

el_args: 1.24.

el_common: 1.24.

el_num: 1.24.

eof: 1.2, 1.4, 1.5, 1.6, 1.7, 1.8.

equivalents: 1.11, 1.13, 1.14, 1.18, 1.20, 1.24, 1.25, 1.26.

file: 1.2.

FILE: [1](#).

FILELEN: 1.2.

filenames_array: 1.2.

finish_var0_list: 1.1, [1.10](#).

first_product: [1.25](#), [1.26](#).

float_abort: 1.1.

form: 1.2.

FORTTRAN77: 1.1.

generic: 1.11.

generic_prod: 1.24.

generic_reagents: 1.11.

generic_1: 1.25, 1.26.

generic_2: 1.25, 1.26.

generic_3: 1.26.

generics: 1.3, 1.4, 1.5, 1.11, 1.13, 1.14, 1.18, 1.20, 1.24, 1.25, 1.26.

i: [1.3](#), [1.4](#), [1.5](#), [1.7](#), [1.8](#), [1.9](#), [1.10](#), [1.11](#), [1.12](#), [1.13](#), [1.14](#), [1.16](#), [1.18](#), [1.19](#), [1.20](#), [1.21](#), [1.24](#), [1.25](#), [1.26](#).

i.e.rate: [1.10](#).

ic: [1.24](#).

implicit_none_f97: 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 1.10, 1.11, 1.12, 1.13, 1.14, 1.15, 1.16,

1.17, 1.18, 1.19, 1.20, 1.21, 1.22, 1.23, 1.24, 1.25, 1.26.

implicit_none_f90: 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 1.10, 1.11, 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.19, 1.20, 1.21, 1.22, 1.23, 1.24, 1.25, 1.26.

init_pmi: 1.1.

init_problem: 1.2, [1.3](#).

init_reaction: 1.1.

init_var0_list: 1.1, [1.9](#).

int_lookup: 1.11.

int_unused: 1.11, 1.19.

j: [1.11](#), [1.24](#), [1.25](#), [1.26](#).

k: [1.11](#), [1.26](#).

l: [1.11](#).

len: 1.2, 1.4, 1.5, 1.6, 1.7, 1.8.

length: [1.2](#), [1.4](#), [1.5](#), [1.6](#), [1.7](#), [1.8](#).

line: [1.2](#), [1.4](#), [1.5](#), [1.6](#), [1.7](#), [1.8](#), [1.10](#).

LINELEN: 1.2.

lines: [1.10](#).

loop1: 1.2, 1.4, 1.5, 1.6, 1.7, 1.8, 1.14.

loop2: 1.4, 1.5, 1.6, 1.7, 1.8.

m: [1.11](#).

ma_common: 1.7, 1.11.

ma_lookup: 1.7.

ma_name: 1.11.

ma_num: 1.7.

match: [1.24](#).

max: 1.11.

max_lines: [1.10](#).

mem_alloc: 1.9.

mem_size: 1.9.

nc_read_elements: 1.3.

nc_read_materials: 1.3.

nc_read_pmi: 1.3.

nc_read_reactions: 1.3.

nc_read_species: 1.3.

nc_write_problem: 1.1.

next_token: 1.4, 1.5, 1.6, 1.7, 1.8.

num_electrons: [1.14](#).

num_equiv: 1.11, 1.13, 1.14, 1.18, 1.20, 1.24, 1.25, 1.26.

num_generics: 1.3, 1.4, 1.5, 1.11, 1.24.

num_lines: [1.10](#).

num_sp: [1.25](#), [1.26](#).

old_num: [1.10](#).

one: 1.11, 1.13, 1.14, 1.15, 1.18, 1.22, 1.23.

p: [1.4](#), [1.5](#), [1.6](#), [1.7](#), [1.8](#).

parse_string: 1.4, 1.5, 1.6, 1.7, 1.8.

pm_args: 1.11, 1.21.
pm_common: 1.8, 1.11, 1.21, 1.22, 1.23, 1.24.
pm_decl: 1.21, 1.22, 1.23, 1.24.
pm_dummy: 1.21, 1.22, 1.23, 1.24.
pm_gen: 1.11.
pm_lookup: 1.8.
pm_name: 1.11.
pm_num: 1.8.
pm_pmi_type: 1.21.
pm_product: 1.11, 1.22, 1.24.
pm_product_num: 1.11, 1.22, 1.23, 1.24.
pm_reagent: 1.11.
pmi_generic_no: 1.11.
pmi_generic_yes: 1.11.
pr: 1.11.
pr_arrangement_max: 1.11, 1.25, 1.26.
pr_background: 1.5, 1.9, 1.11.
pr_background_check: 1.11.
pr_background_decl: 1.11.
pr_background_lookup: 1.11.
pr_background_num: 1.3, 1.5, 1.9, 1.11.
pr_bk_rc: 1.11, 1.19, 1.20.
pr_bkrc_dim: 1.11.
pr_bkrc_num: 1.11.
pr_bkrc_prod: 1.11, 1.19, 1.20.
pr_bkrc_reagent_max: 1.11.
pr_bkrc_rg: 1.11.
pr_common: 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 1.10, 1.11, 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.19, 1.20, 1.21, 1.22, 1.23, 1.24, 1.25, 1.26.
pr_generic_decl: 1.11, 1.24, 1.25, 1.26.
pr_mat_ref: 1.7, 1.11.
pr_materials_num: 1.3, 1.7, 1.11.
pr_max_equiv: 1.11.
pr_num_arrangements: 1.11, 1.12, 1.13, 1.14, 1.15, 1.16, 1.18, 1.25, 1.26.
pr_num_change_vars: 1.9.
pr_pm_case_num: 1.11, 1.21, 1.22, 1.23, 1.24.
pr_pm_cases: 1.11.
pr_pm_num_arrange: 1.11, 1.21, 1.22, 1.23, 1.24.
pr_pm_prod: 1.11, 1.22, 1.23, 1.24.
pr_pm_prod_mult: 1.11, 1.21, 1.22, 1.23, 1.24.
pr_pm_ref: 1.8, 1.11.
pr_pmi_max: 1.11.
pr_pmi_num: 1.3, 1.8, 1.11.
pr_problem_sp_back: 1.9.
pr_problem_sp_test: 1.9.
pr_prod_mult: 1.11, 1.12, 1.13, 1.14, 1.15, 1.18, 1.25, 1.26.
pr_rc_num: 1.11, 1.12, 1.13, 1.14, 1.15, 1.16, 1.18, 1.25, 1.26.
pr_reaction: 1.6, 1.11, 1.19, 1.20.

pr_reaction_dim: 1.6.
pr_reaction_max: 1.11.
pr_reaction_num: 1.3, 1.6, 1.11.
pr_tag_string_length: 1.10.
pr_test: 1.4, 1.9, 1.11, 1.22, 1.24.
pr_test_args: 1.11, 1.12, 1.16, 1.17, 1.21.
pr_test_check: 1.11.
pr_test_decl: 1.11, 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.21, 1.22, 1.23, 1.24, 1.25, 1.26.
pr_test_dummy: 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.21, 1.22, 1.23, 1.24, 1.25, 1.26.
pr_test_lookup: 1.11.
pr_test_num: 1.3, 1.4, 1.9, 1.11.
pr_ts_bk: 1.11.
pr_ts_prod: 1.11, 1.13, 1.14, 1.15, 1.16, 1.18, 1.25, 1.26.
pr_ts_rc: 1.11.
pr_var_angle: 1.9.
pr_var_emitter_v_vector: 1.9.
pr_var_emitter_v_2: 1.9.
pr_var_emitter_v_3: 1.9.
pr_var_emitter_vf_Maxwell_vector: 1.9.
pr_var_emitter_vf_Maxwell_2: 1.9.
pr_var_emitter_vf_Maxwell_3: 1.9.
pr_var_emitter_vth_Maxwell: 1.9.
pr_var_energy: 1.9.
pr_var_energy_change: 1.9.
pr_var_energy_in: 1.9.
pr_var_energy_out: 1.9.
pr_var_mass: 1.9.
pr_var_mass_change: 1.9.
pr_var_mass_in: 1.9.
pr_var_mass_out: 1.9.
pr_var_momentum_change_vector: 1.9.
pr_var_momentum_change_2: 1.9.
pr_var_momentum_change_3: 1.9.
pr_var_momentum_in_vector: 1.9.
pr_var_momentum_in_2: 1.9.
pr_var_momentum_in_3: 1.9.
pr_var_momentum_out_vector: 1.9.
pr_var_momentum_out_2: 1.9.
pr_var_momentum_out_3: 1.9.
pr_var_momentum_vector: 1.9.
pr_var_momentum_2: 1.9.
pr_var_momentum_3: 1.9.
pr_var_problem_sp_index: 1.9.
pr_var_unknown: 1.9.
pr_var_xz_stress: 1.9.
pr_var0_list: 1.9, 1.10.
pr_var0_num: 1.9, 1.10.
problem_background_reaction: 1.11.
problem_background_sp: 1.5, 1.11.

- problem_bkrc_products*: 1.11.
- problem_bkrc_reagents*: 1.11.
- problem_infile*: 1.2.
- problem_materials_ref*: 1.7, 1.11.
- problem_materials_sub*: 1.7.
- problem_num_arrangements*: 1.11.
- problem_pmi_case_num*: 1.11.
- problem_pmi_cases*: 1.11.
- problem_pmi_num_arrange*: 1.11.
- problem_pmi_prod_mult*: 1.11.
- problem_pmi_products*: 1.11.
- problem_pmi_ref*: 1.8, 1.11.
- problem_pmi_sub*: 1.8.
- problem_prod_mult*: 1.11.
- problem_rc*: 1.6, 1.11.
- problem_reaction_num*: 1.11.
- problem_species_background*: 1.5.
- problem_species_test*: 1.4.
- problem_test_background*: 1.11.
- problem_test_products*: 1.11.
- problem_test_reaction*: 1.11.
- problem_test_sp*: 1.4, 1.11.
- problem_version*: 1.2.
- problemfile*: 1.
- problemsetup*: 1, 1.1.
- product*: 1.13, 1.14, 1.18, 1.20, 1.25, 1.26.
- product_perm_2*: 1.16, 1.17, 1.25.
- product_perm_3*: 1.16, 1.17, 1.26.
- ps_common*: 1.3, 1.4, 1.5, 1.11, 1.13, 1.14, 1.18, 1.20, 1.24, 1.25, 1.26.
- rc_args*: 1.11, 1.12, 1.16, 1.17.
- rc_common*: 1.6, 1.11, 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.19, 1.20, 1.25, 1.26.
- rc_decl*: 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.20, 1.25, 1.26.
- rc_dummy*: 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.25, 1.26.
- rc_gen*: 1.11.
- rc_generic_no*: 1.11.
- rc_generic_yes*: 1.11.
- rc_lookup*: 1.6.
- rc_name*: 1.11.
- rc_product*: 1.11, 1.13, 1.14, 1.15, 1.16, 1.18, 1.20, 1.25, 1.26.
- rc_product_max*: 1.11, 1.19.
- rc_product_num*: 1.11, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.20.
- rc_reaction_type*: 1.12, 1.19.
- rc_reagent*: 1.11.
- rc_reagent_max*: 1.11.
- rc_reagent_num*: 1.11.
- read_background*: 1.2, 1.5.
- read_materials*: 1.2, 1.7.
- read_pmi*: 1.2, 1.8.
- read_problem*: 1.1, 1.2.
- read_reaction*: 1.2, 1.6.
- read_string*: 1.2, 1.4, 1.5, 1.6, 1.7, 1.8.
- read_test*: 1.2, 1.4.
- readfilenames*: 1.1.
- reagents*: 1.11, 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.19, 1.20.
- real_unused*: 1.11.
- rf_common*: 1.2.
- set_bkrc_products*: 1.11, 1.19.
- set_pmi_products*: 1.11, 1.21.
- set_prod_adsorption*: 1.21, 1.23.
- set_prod_charger*: 1.12, 1.14.
- set_prod_desorption*: 1.21, 1.24.
- set_prod_dissoc*: 1.12, 1.16.
- set_prod_dissoc_rec*: 1.12, 1.17.
- set_prod_elastic*: 1.12, 1.15.
- set_prod_excite*: 1.12, 1.18.
- set_prod_ionize*: 1.12, 1.13.
- set_prod_recombine*: 1.19, 1.20.
- set_prod_reflection*: 1.21, 1.22.
- set_products*: 1.11, 1.12.
- sp*: 1.25, 1.26.
- SP*: 1.10.
- sp_args*: 1.11, 1.12, 1.13, 1.14, 1.16, 1.17, 1.18, 1.19, 1.20, 1.25, 1.26.
- sp_common*: 1.3, 1.4, 1.5, 1.9, 1.11, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.20, 1.22, 1.25, 1.26.
- sp_decl*: 1.11, 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.19, 1.20, 1.25, 1.26.
- sp_dummy*: 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.19, 1.20, 1.25, 1.26.
- sp_generic*: 1.4, 1.5, 1.13, 1.15, 1.16, 1.18, 1.22.
- sp_lookup*: 1.4, 1.5, 1.13, 1.14.
- sp_m*: 1.11.
- sp_multiplicity*: 1.25, 1.26.
- sp_name*: 1.11.
- sp_num*: 1.3, 1.4, 1.5.
- sp_sy*: 1.9.
- species_add_check*: 1.13, 1.14, 1.15, 1.18, 1.20, 1.25, 1.26.
- species_el_count*: 1.24.
- st_decls*: 1.2, 1.4, 1.5, 1.6, 1.7, 1.8, 1.10, 1.11, 1.13, 1.14, 1.15, 1.18, 1.20.
- status**: 1.2.
- stdout*: 1.11.
- string_lookup*: 1.10.
- sum*: 1.12, 1.21.
- SUN*: 1.1.

tempfile: 1.2.
test: 1.11.
trim: 1.11.
ts: 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.21, 1.22,
1.23, 1.24, 1.25, 1.26.
ts_reagent: 1.11.
unit: 1.2, 1.4, 1.5, 1.6, 1.7, 1.8.
var_alloc: 1.3, 1.4, 1.5, 1.7, 1.8, 1.9, 1.11.
var_realloc: 1.9, 1.10.
var_realloca: 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 1.10, 1.11.
var_reallocb: 1.10, 1.11.
xs_copy: 1.9, 1.10.
zero: 1.12, 1.21, 1.24.

⟨ Functions and Subroutines 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 1.10, 1.11, 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.19, 1.20, 1.21, 1.22, 1.23, 1.24, 1.25, 1.26 ⟩ Used in section 1.1.
⟨ Memory allocation interface 0 ⟩ Used in sections 1.11, 1.10, 1.9, 1.8, 1.7, 1.6, 1.5, 1.4, 1.3, and 1.2.

COMMAND LINE: "fweave -f -i! -W[-ykw800 -ytw40000 -j -n/
/u/dstotler/degas2/src/problemsetup.web".

WEB FILE: "/u/dstotler/degas2/src/problemsetup.web".

CHANGE FILE: (none).

GLOBAL LANGUAGE: FORTRAN.